

CS173: Discrete Structures

Problem Set

Mohammed Al Salhi

1. Prove $(\forall x \in \mathbb{N})(S(x) \neq \emptyset)$.

Proof. Let $n \in \mathbb{N}$. We proceed by contradiction. Assume $S(n) = \emptyset$.

By definition, $S(n) = n \cup \{n\}$, so we have:

$$n \cup \{n\} = \emptyset.$$

But $n \in \{n\}$, so $n \in n \cup \{n\}$, which means $n \cup \{n\} \neq \emptyset$ is a contradiction.

Therefore $S(n) \neq \emptyset$ for all $n \in \mathbb{N}$. □

2. Prove $(\forall n \in \mathbb{N})(n \neq 0 \Rightarrow (\exists m \in \mathbb{N})(S(m) = n))$.

Proof. Let $n \in \mathbb{N}$ and assume $n \neq 0$.

By the definition of natural numbers, every natural number is either 0 or the successor of some natural number. Since $n \neq 0$, it must be the case that $n = S(m)$ for some $m \in \mathbb{N}$.

This m is our witness and $S(m) = n$. □

3. Prove $(\forall n \in \mathbb{N})(\forall m \in \mathbb{N})(S(n + m) = S(n) + m)$.

Proof. Let $n \in \mathbb{N}$. We induct on m .

Base case: $m = 0$. We need to show $S(n + 0) = S(n) + 0$.

$$S(n + 0) = S(n) = S(n) + 0$$

where the first equality is by definition of addition and the second is by definition of addition.

Inductive hypothesis: Assume $S(n + k) = S(n) + k$ for some $k \in \mathbb{N}$.

Inductive step: We need to show $S(n + S(k)) = S(n) + S(k)$.

$$S(n + S(k)) = S(S(n + k)) = S(S(n) + k) = S(n) + S(k)$$

where the first equality is by definition of addition, the second is by the inductive hypothesis, and the third is by definition of addition. □

4. Prove $(\forall n \in \mathbb{N})(0 \leq n)$.

Proof by induction:

Proof. We induct on n .

Base case: $n = 0$. We need to show $0 \leq 0$, which holds immediately since $\leq$ is reflexive.

Inductive hypothesis: Assume $0 \leq k$ for some $k \in \mathbb{N}$.

Inductive step: We need to show $0 \leq S(k)$. By the inductive hypothesis $0 \leq k$, and by definition $k < S(k)$, so by transitivity $0 \leq S(k)$. $\square$

Proof by definition of $\leq$:

Proof. Let $n \in \mathbb{N}$. By definition, $0 \leq n$ if and only if there exists $m \in \mathbb{N}$ such that $0 + m = n$. Taking $m = n$, we have $0 + n = n$ by definition of addition. Hence $0 \leq n$. $\square$

5. Prove $(\forall n \in \mathbb{N})(n \neq 0 \Rightarrow 0 < n)$.

Proof. We induct on n .

Base case: NTS: $0 \neq 0 \Rightarrow 0 < 0$. Assume $0 \neq 0$. We know $(\forall x)(x = x)$, so $0 = 0$ — a contradiction. Thus the implication holds vacuously.

Inductive hypothesis: Assume $k \neq 0 \Rightarrow 0 < k$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $S(k) \neq 0 \Rightarrow 0 < S(k)$. Assume $S(k) \neq 0$. By definition of $<$, we have $0 < S(k)$ if and only if $0 \leq k$, which holds by the previous theorem. $\square$

6. Prove $(\forall x \in \mathbb{N})(\forall y \in \mathbb{N})(x + y = y + x)$.

Lemma: $(\forall n \in \mathbb{N})(\forall x \in \mathbb{N})(S(n) + x = S(n + x))$.

Proof. Let $n \in \mathbb{N}$. We induct on x .

Base case: NTS: $S(n) + 0 = S(n + 0)$.

$$S(n) + 0 = S(n) = S(n + 0)$$

where the first equality is by definition of addition and the second is by definition of addition.

Inductive hypothesis: Assume $S(n) + k = S(n + k)$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $S(n) + S(k) = S(n + S(k))$.

$$S(n) + S(k) = S(S(n) + k) = S(S(n + k)) = S(n + S(k))$$

where the first equality is by definition of addition, the second is by the inductive hypothesis, and the third is by definition of addition. $\square$

Main proof:

Proof. Let $x \in \mathbb{N}$. We induct on y .

Base case: NTS: $x + 0 = 0 + x$. By definition of addition, $x + 0 = x$. By theorem, $0 + x = x$. Hence $x + 0 = 0 + x$.

Inductive hypothesis: Assume $x + k = k + x$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $x + S(k) = S(k) + x$.

$$x + S(k) = S(x + k) = S(k + x) = S(k) + x$$

where the first equality is by definition of addition, the second is by the inductive hypothesis, and the third is by the lemma. $\square$

7. Prove $(\forall x \in \mathbb{N})((x \cdot 1 = x) \wedge (1 \cdot x = x))$.

Proof. We induct on x .

Base case: NTS: $(0 \cdot 1 = 0) \wedge (1 \cdot 0 = 0)$.

Consider $0 \cdot 1$:

$$0 \cdot 1 = 0 \cdot S(0) = (0 \cdot 0) + 0 = 0 + 0 = 0$$

where the first equality is by definition of $1 = S(0)$, the second is by definition of multiplication, and the third and fourth are by definition of addition.

Consider $1 \cdot 0$:

$$1 \cdot 0 = 0$$

by definition of multiplication.

Inductive hypothesis: Assume $k \cdot 1 = k$ and $1 \cdot k = k$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $S(k) \cdot 1 = S(k)$ and $1 \cdot S(k) = S(k)$.

Consider $S(k) \cdot 1$:

$$S(k) \cdot 1 = S(k) \cdot S(0) = (S(k) \cdot 0) + S(k) = 0 + S(k) = S(k)$$

where the first equality is by definition of $1 = S(0)$, the second is by definition of multiplication, the third is by definition of multiplication, and the fourth is by theorem.

Consider $1 \cdot S(k)$:

$$1 \cdot S(k) = (1 \cdot k) + 1 = k + 1 = S(k)$$

where the first equality is by definition of multiplication, the second is by the inductive hypothesis, and the third is by definition of $S(k) = k + 1$. $\square$

8. Prove $(\forall x \in \mathbb{N})((x \cdot 0 = 0) \wedge (0 \cdot x = 0))$.

Proof. For the first conjunct, $x \cdot 0 = 0$ holds directly by definition of multiplication.

For the second conjunct, we induct on x to prove $(\forall x \in \mathbb{N})(0 \cdot x = 0)$.

Base case: NTS: $0 \cdot 0 = 0$. This holds by definition of multiplication.

Inductive hypothesis: Let $k \in \mathbb{N}$ and assume $0 \cdot k = 0$.

Inductive step: NTS: $0 \cdot S(k) = 0$.

$$0 \cdot S(k) = (0 \cdot k) + 0 = 0 + 0 = 0$$

where the first equality is by definition of multiplication, the second is by the inductive hypothesis, and the third is by definition of addition. $\square$

9. Prove $(\forall x, y, z \in \mathbb{N})(x \cdot (y + z) = (x \cdot y) + (x \cdot z))$.

Proof. Let $x, y \in \mathbb{N}$. We induct on z to prove $(\forall z \in \mathbb{N})(x \cdot (y + z) = (x \cdot y) + (x \cdot z))$.

Base case: NTS: $x \cdot (y + 0) = (x \cdot y) + (x \cdot 0)$.

Consider the LHS:

$$x \cdot (y + 0) = x \cdot y$$

by definition of addition. Consider the RHS:

$$(x \cdot y) + (x \cdot 0) = (x \cdot y) + 0 = x \cdot y$$

where the first equality is by definition of multiplication and the second is by definition of addition. Hence LHS = RHS.

Inductive hypothesis: Assume $x \cdot (y + k) = (x \cdot y) + (x \cdot k)$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $x \cdot (y + S(k)) = (x \cdot y) + (x \cdot S(k))$.

$$\begin{aligned} x \cdot (y + S(k)) &= x \cdot S(y + k) = (x \cdot (y + k)) + x = ((x \cdot y) + (x \cdot k)) + x \\ &= (x \cdot y) + ((x \cdot k) + x) = (x \cdot y) + (x \cdot S(k)) \end{aligned}$$

where the first equality is by definition of addition, the second is by definition of multiplication, the third is by the inductive hypothesis, the fourth is by associativity of addition, and the fifth is by definition of multiplication. $\square$

10. Prove $(\forall x, y \in \mathbb{N})(x \cdot y = y \cdot x)$.

Lemma: $(\forall k \in \mathbb{N})(\forall x \in \mathbb{N})(S(k) \cdot x = (k \cdot x) + x)$.

Proof. Let $k \in \mathbb{N}$. We induct on x .

Base case: NTS: $S(k) \cdot 0 = (k \cdot 0) + 0$.

$$S(k) \cdot 0 = 0 = 0 + 0 = (k \cdot 0) + 0$$

where the first equality is by definition of multiplication, the second is by definition of addition, and the third is by definition of multiplication.

Inductive hypothesis: Assume $S(k) \cdot x = (k \cdot x) + x$ for some $x \in \mathbb{N}$.

Inductive step: NTS: $S(k) \cdot S(x) = (k \cdot S(x)) + S(x)$.

$$\begin{aligned} S(k) \cdot S(x) &= (S(k) \cdot x) + S(k) = ((k \cdot x) + x) + S(k) \\ &= (k \cdot x) + (x + S(k)) = (k \cdot x) + S(x + k) \\ &= (k \cdot x) + S(k + x) = (k \cdot x) + (k + S(x)) \\ &= (k \cdot S(x)) + S(x) \end{aligned}$$

where the first equality is by definition of multiplication, the second is by the inductive hypothesis, the third and fourth are by associativity and definition of addition, the fifth is by commutativity of addition, the sixth is by definition of addition, and the seventh is by definition of multiplication. $\square$

Main proof:

Proof. Let $x \in \mathbb{N}$. We induct on y .

Base case: NTS: $x \cdot 0 = 0 \cdot x$. By definition of multiplication, $x \cdot 0 = 0$. By theorem, $0 \cdot x = 0$. Hence $x \cdot 0 = 0 \cdot x$.

Inductive hypothesis: Assume $x \cdot k = k \cdot x$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $x \cdot S(k) = S(k) \cdot x$.

$$x \cdot S(k) = (x \cdot k) + x = (k \cdot x) + x = S(k) \cdot x$$

where the first equality is by definition of multiplication, the second is by the inductive hypothesis, and the third is by the lemma. $\square$

11. Prove $(\forall x, y, z \in \mathbb{N})(x^y \cdot x^z = x^{y+z})$.

Proof. Let $x, y \in \mathbb{N}$. We induct on z .

Base case: NTS: $x^y \cdot x^0 = x^{y+0}$.

$$x^y \cdot x^0 = x^y \cdot 1 = x^y = x^{y+0}$$

where the first equality is by definition of exponentiation, the second is by definition of multiplication, and the third is by definition of addition.

Inductive hypothesis: Assume $x^y \cdot x^k = x^{y+k}$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $x^y \cdot x^{S(k)} = x^{y+S(k)}$.

$$x^y \cdot x^{S(k)} = x^y \cdot (x^k \cdot x) = (x^y \cdot x^k) \cdot x = x^{y+k} \cdot x = x^{S(y+k)} = x^{y+S(k)}$$

where the first equality is by definition of exponentiation, the second is by associativity of multiplication, the third is by the inductive hypothesis, the fourth is by definition of exponentiation, and the fifth is by definition of addition. $\square$

12. Prove $(\forall n \in \mathbb{N}) (6 \cdot \sum_{i=0}^n i^2 = n(n+1)(2n+1))$.

Proof. We induct on n .

Base case: NTS: $6 \cdot \sum_{i=0}^0 i^2 = 0 \cdot 1 \cdot 1$.

$$6 \cdot \sum_{i=0}^0 i^2 = 6 \cdot 0 = 0 = 0 \cdot 1 \cdot 1$$

as required.

Inductive hypothesis: Assume $6 \cdot \sum_{i=0}^k i^2 = k(k+1)(2k+1)$ for some $k \in \mathbb{N}$.

Inductive step: NTS: $6 \cdot \sum_{i=0}^{S(k)} i^2 = S(k)(S(k)+1)(2S(k)+1)$.

$$\begin{aligned} 6 \cdot \sum_{i=0}^{S(k)} i^2 &= 6 \cdot \left(\sum_{i=0}^k i^2 + S(k)^2 \right) = 6 \cdot \sum_{i=0}^k i^2 + 6 \cdot S(k)^2 \\ &= k(k+1)(2k+1) + 6(k+1)^2 \\ &= (k+1) [k(2k+1) + 6(k+1)] \\ &= (k+1)(2k^2 + k + 6k + 6) \\ &= (k+1)(2k^2 + 7k + 6) \\ &= (k+1)(k+2)(2k+3) \\ &= S(k)(S(k)+1)(2S(k)+1) \end{aligned}$$

where the first equality splits the sum, the second distributes the 6, the third applies the inductive hypothesis and uses $S(k) = k+1$, and the remaining steps are algebraic simplification. $\square$